

# ZS-M47

## The Seam Modular-Depth Theorem: Faithful Genuineness, the Two-Leg Clock Gate, and the Compact-Dual Weyl-3 Dirac Operator

*Program Theorem 2 of 3 of the ZS-A31 §6.3 Modular-Depth Line —  
Delivered Conditionally on ZS-M46's Own Registry*

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**Theme / Code:** Math Spine / ZS-M47 v2.0 (Final) · Downstream of ZS-M46 / ZS-M1 / ZS-M2 / ZS-A17 / ZS-A23 / ZS-A24 / ZS-A30 / ZS-F23 / ZS-F30 / ZS-F31 / ZS-F32 · Upstream of ZS-F37 / ZS-A32

**Verification:** exact/numerical checks **148/148 PASS** [v1.0–v1.9 suites 139, all re-run + v2.0 additions 9: GEO 3 | CY 5 | CN 1]; executed consistency guards **13/13 pass**; **17 imported-theorem conditions registered, NOT machine-certified; anti-numerology MC executed (CY4:  $p = 0.352$  — non-identifications confirmed correct)**. Companion scripts accompany this release. **Zero fitted parameters**;  $(A, Q, \dim Z) = (35/437, 11, 2)$  LOCKED.

**Scripts:** zs\_m47\_verify\_v1\_0.py ... zs\_m47\_verify\_v2\_0\_additions.py (SymPy + mpmath 50-digit + NumPy).

**Retraction notice (permanent record).** Theorem M47.P and Corollary CRT-1' of v1.0/v1.1 — the direct Araki–Woods classification of  $\otimes(M_3 \oplus \mathbb{C} \oplus M_5)$  — are **RETRACTED** (the local algebra is not a factor; the block-tracial state is a trace; machine-confirmed PE1) and superseded by the Parent-Factor Realization Problem (§5).

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## §0. Abstract

This is the final version of the second of the three ZS-A31 §6.3 program theorems. Its charter — the metric-dimension-3 Dirac operator (NDC3) built from the Z-Spin seam's modular data — is delivered, conditionally, together with the two structures the delivery required: a complete faithful-genuineness theory of the seam inclusion and a decided clock gate. v2.0 executes the three closing follow-ups — (a) the (SZ-geo) decision, (b) the final consolidation, (c) the period-3 structure with an executed anti-numerology Monte Carlo — and states the single delivery theorem.

**Theorem M47 (the delivery theorem; the full statement is §12).** *Let  $(N \subset M, \Omega)$  be the Z-Spin seam inclusion — the half-sided modular inclusion generated by the ZS-M1 Koenigs germ under ZS-M46's conditions KH1–KH4. Then:*

**(I) Canonical structure [DERIVED].** *The canonical endomorphism is the doubled seam step,  $\Gamma = J_N J_M = U(2)$ ; the Jones–Longo tunnel is the ZS-M46 T6 depth chain; the Wiesbrock intersection datum is  $U(1)$ ; translations are never inner ( $U(a) \notin M$ ).*

**(II) Faithful genuineness [DERIVED-CONDITIONAL, model-fixed].** *No faithful normal conditional expectation  $M \rightarrow N$  exists in the strongly-additive chiral realization; equivalently  $(\star_s)$  holds. T4- $\star$  reduces to the charged-intertwiner statement  $(\star\star)$ , fenced by seven proven constraints, with its analytic core — and simultaneously ZS-M46's KH4 — secured by the **Szegő gate**: the seam dynamics is critical-point-free (its derivative is proportional to itself), the boundary decay rate is the exact Julia-anchor constant  $a = \ln(GM|_{\text{cycle}}/|z^*|) = 0.46274971$ , and the strip-end identification is the defining intertwining property of the suspension — so the gate rests on **KH1–KH4 alone**.*

**(III) The two-leg clock gate [DERIVED-CONDITIONAL on (MX) + (H1)].** *The register transport is the gauge-invariant two-leg character law: the corner weight is the size-biased  $\omega = (4, 9, 36)/49$ , exactly at vanishing seam rapidity and within 1% at ZS-F30's rapidity bound; on this branch the E1 parent envelope is the injective type III<sub>1</sub> factor  $R_\infty$  and CRT-4a survives (the no-mixed-outcome theorem). Every representation input is corpus-anchored:  $X$  carries F30's spin-1 axes,  $Z$  carries F23's spin- $\frac{1}{2}$  grading ( $\beta =$  the  $2\pi$ -rotation sign, the deck involution in half-angle coordinates), and  $Y \cong Z \otimes X$  is **selected by the founding mediation axiom itself** — Clebsch–Gordan arithmetic kills every mediation vertex of the alternative.*

**(IV) NDC3 — the charter [DERIVED-CONDITIONAL on KH1–KH4 + (H-2D) + (H-cd)].** *The conjugate Koenigs seam exists under the same KH registry (the conjugation-transfer theorem); the tensor two-seam net is a 1+1 Borchers triple whose symmetry  $\text{Möb} \times \text{Möb}$  is the isometry group of  $\text{AdS}_3 =$  the  $SL(2, \mathbb{R})$  group manifold (Rehren holography: a 2+1 wedge net); and on the compact dual  $SU(2) \cong S^3$  — **the unit quaternions** — the Dirac operator has the exact spectrum  $\pm(n+3/2)$  with multiplicity  $(n+1)(n+2)$  and the closed-form cubic counting  $N(\lambda) \sim (2/3)\lambda^3$ : **the metric-dimension-3 Weyl law of ZS-A17 Theorem F, exactly**. The boundary alternative is closed by the Cardy obstruction (partition-multiplicity growth), so the compact-dual placement is a necessity, not a choice.*

**v2.0's three closings.** (a) **(SZ-geo) discharged [DERIVED]:** any KH suspension intertwines the dynamics with a real modular translation,  $s(f(z)) = s(z) + c_0$ ; hence forward orbits go to one strip end and backward orbits to the other **by definition of the construction** (GEO1) — verified mechanically by the branch-consistent Abel increment  $u(fz) - u(z) = 1$  with independently computed Koenigs values at thirty orbit points (max deviation  $3.3 \times 10^{-15}$ ; GEO2) and the exact branch period  $\tau = 2\pi i / \log \lambda^* = 2.7743 - 0.1410i$  (GEO3). The Szegő gate's condition registry collapses to KH1–KH4. (b) **The consolidation:** the delivery theorem above; the condition-trajectory table (§12.2) documenting four discharged hypothesis chains, one retraction, and the final registry of eight named conditions (four of which are ZS-M46's own) plus four open mathematical items; the ZS-F37/ZS-A32 handover table (§12.3). (c) **The period-3**

**structure, with the anti-numerology MC executed:** the Julia anchor's three points have moduli (1.3413, 1.6574, 0.3296) — a genuine period-3 orbit of the entire map,  $f^3(z_1) = z_1$  to fifty digits (CY1) — with the multiplier identity  $|(f^3)'| = (\pi/2)^3 \prod |z_i| = 2.8398$  exact (CY2); the nearest-constant scan finds **no corpus match** for  $a = 0.46275$  or  $GM = 0.90152$  at relative  $10^{-3}$  (CY3); the Monte Carlo returns  $p = 0.352 \gg 5\%$  — the observed proximities are unremarkable and the non-identifications are confirmed correct (CY4); and the period is stable under  $\pm 2\%$  and  $\pm 5\%$  perturbations of the exponent coefficient (CY5) — **the "3" is a structurally stable dynamical feature, not a corpus fingerprint:** the period-3/dim-X coincidence-candidate remains firewalled with quantitative justification.

Zero fitted parameters throughout; every number in this paper is a locked input, a classical constant, or an exact evaluation of the locked dynamics.

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## Epistemic Status Legend

| Tag                       | Meaning                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------|
| PROVEN                    | Explicit proof or exact verification; no undischarged assumption.                                     |
| IMPORTED-PROVEN           | Proven in the external literature and used without re-proof; cited.                                   |
| DERIVED                   | Follows from PROVEN / IMPORTED-PROVEN results by stated steps; no new parameter.                      |
| DERIVED-CONDITIONAL       | Derived modulo explicitly named, falsifiable conditions.                                              |
| OBSTRUCTION               | A proven negative result: a candidate object, method, or arena fails a required property.             |
| HYPOTHESIS-weak / -strong | A registered candidate; strong when consistency checks pass and anti-numerology MC is pre-registered. |
| OPEN                      | A genuine, well-posed gap; may carry a finite-decision reduction.                                     |
| NON-CLAIM                 | Explicitly outside this paper's scope (deferred to a named successor).                                |
| RETRACTED                 | A previously published claim withdrawn with stated reason; superseded text identified.                |
| CONSISTENCY-CHECK         | A numerical, character, model, chronology, or scope check that does NOT by itself certify a theorem.  |

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# §1. Introduction

The ZS-A31 §6.3 program charged three papers with converting the Z-Spin seam's modular data into geometry. This paper — the second — was charged with the depth line: the canonical structure of the seam inclusion and, from it, a metric-dimension-3 Dirac operator. Ten versions later, the delivery stands as Theorem M47 (§12), and the path there is itself part of the result: one retraction honestly executed and rebuilt into a frontier problem (the Parent-Factor Realization Problem); four hypothesis chains discharged into corpus text, definitional identities, or arithmetic (the parity grading, the representation designation, the Szegő tail, the strip-end geometry); two direct routes converted into proven obstructions that forced the surviving one (the Lorentzian Dirac; the boundary spectra); and a condition registry that ends *shorter* than it began, with half of it being ZS-M46's own KH set.

The recurring pattern deserves its one-sentence statement: **at every decision point, the answer was already inside the corpus** — the transport's parity is the definition of the support-ratio rapidity; the Y-sector's representation is forced by the founding sentence "X and Y exchange information through Z"; the third dimension is the seam's own Möbius group; the tail rate is the modulus of the Julia anchor the seam's own inverse branch finds; and the strip-end identification is the intertwining property that defines a suspension at all.

**Terminology.** The mediating *action* is **Z-Spin mediation** (the actor is the Spin); **Z-sector** denotes the geometric stage; hence Z-Spin-mediated coupling, Z-Spin dynamics, Z-Spin iteration throughout. The micro triad: X-sector  $\leftrightarrow$  particle, Y-sector  $\leftrightarrow$  wave, Z-sector  $\leftrightarrow$  Spin.

**Locked data (ZS-M1, ZS-F31 App. D).**  $A = 35/437$ ,  $Q = 11$ ,  $(\dim Z, \dim X, \dim Y) = (2, 3, 6)$ ;  $\kappa^2 = A/Q = 35/4807$ ;  $z^* = 0.43828 + 0.36059i$ ;  $\lambda^* = (i\pi/2)z^*$ ,  $|\lambda^*| = 0.89151$ ,  $\mu = 0.1148346250$ ,  $\theta = 2.2592495540$  (K1–K4). Sector order (Z, X, Y) (ORD1–ORD2).

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## §2. Setting and the final condition registry

$(N \subset M, \Omega)$  standard HSMI; Wiesbrock/Araki–Zsidó structure  $U(a) = e^{\{iaP\}}$ ,  $P \geq 0$ ,  $N = U(1)MU(-1)$  [IMPORTED-PROVEN];  $(UV) \ker P = \mathbb{C}\Omega$ ;  $(\star\_s)/(\star\_w)$  the state/weight invariance statements; **(TT)** tail triviality.

**The final named conditions (eight; CN1): KH1–KH4** (ZS-M46's own registry — the Koenigs-to-standard-pair construction); **(H-2D)** — the physical two-chiral structure is the tensor construction  $H \otimes \bar{H}$ ; **(H-cd)** — the NDC3 spectral frame is the compact dual (necessity-justified by the Cardy obstruction); **(MX)** — the founding mediation axiom: the Z-Spin mediation channel between the X- and Y-sectors is nonvanishing; **(H1)** — corner compatibility of the transported weight.

**The open mathematical items (four): (★★)** beyond its analytic core (the localization residual);  $(\star\_w)$ ;  $P_2$  (the factor envelope); **(TT)** in the abstract setting.

**Discharged chains (four; documented in §12.2):**  $(H-\beta) \rightarrow (H-\beta') \rightarrow$  the T12 textual anchor;  $(H-\text{rep}) \rightarrow$  HR anchoring + YM selection;  $(\text{SZ-tail}) \rightarrow$  the ST tail-rate theorem;  $(\text{SZ-geo}) \rightarrow$  the GEO intertwining argument (this version).

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### §3. Faithful genuineness (consolidated)

**T4-A [PROVEN]:** no faithful normal CE  $\Leftrightarrow (\star\_s)$  (Takesaki, both directions). **T4-B [PROVEN implication]:**  $(\star\_w) \Rightarrow$  no f.n.s. operator-valued weight (Haagerup). **T4-C [PROVEN given (UV)]:** the product-state mechanism. **T4-D [DERIVED-CONDITIONAL, model-fixed]:** no faithful normal CE in the strongly-additive chiral realization (D'Antoni–Longo interpolation + the Möbius element  $g(z) = 1/(1-z) + \text{non-splitness}$ ); establishes  $(\star\_s)$  model-fixed; does not reach operator-valued weights. **T4-E [DERIVED]:**  $\Gamma = J\_N J\_M = U(2)$  (Borchers commutation; GE1–GE2); **T4-E':** the tunnel = the T6 chain. **T4-F:** intertwiner  $\Leftrightarrow$  CE; **T4- $\star \Leftrightarrow (\star\star)$ .**

**The seven-constraint fence on  $(\star\star)$  [all PROVEN except as noted]:** any intertwiner is (1) non-second-quantized (O1-a: cyclicity of the complementary standard subspace); (2) a proper isometry (O1-c); (3) non-translation-commuting (O1-d, under (TT) — the Longo–Witten/Tanimoto class is disjoint); (4) dilation-asymptotically scalar (O1-b) with (5) strictly subunit vacuum overlap (O1-e, via  $U(a) \notin M$ , O1-L); (6) of geometrically decaying ranges with the rate pincer  $|c|^2 \leq \omega(p) < 1$  (O1-g/h/h'); (7) with Hardy-boundary return data  $\text{Im } R(2) = 0$ ,  $\text{Re } R(2) > 0$  in the vacuum-orthogonal corner, where it totally evaporates in Cesàro mean (O1-j/j') — the corner's non-exclusion by these constraints alone is machine-witnessed (OJ1), and its **analytic core is secured by the Szegő gate (§9)**. **O1 abstract closure: OPEN** (localization-essential; the type III Cuntz permissiveness scoping remark stands).

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### §4. The V-C circularity audit

**[CONSISTENCY-CHECK: PASS]** — chronology DAG  $1984 < 1993 < 2022$ ; implication directions audited; the free-case Buchholz witness (C1).

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### §5. The Parent-Factor Realization Problem

**§5.0 [RETRACTED]:** the v1.0/v1.1 direct classification (PE1: the block-tracial state is a trace). **P1 [DERIVED]:** the finite record corner  $\tau\_a = 1/Q$ ,  $\omega\_a = d\_i/49$ ,  $h = Qd/49 = (22, 33, 66)/49$ . **P2 [OPEN]:** the factor envelope (E1 local  $M_{11}$  / E2 tail-ergodic crossed product / E3 central integral). **P3 [DERIVED-CONDITIONAL]:** on E1, the tracial state yields the hyperfinite  $\text{II}_1$  factor (ratio set  $\{1\}$ ; PE2) and the size-biased state yields the injective  $\text{III}_1$  factor  $R_\infty$  (ratios  $\supseteq \{3/2, 2, 3\}$  dense; PE3). **CRT-1: OPEN under parent-factor realization.**

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## §6. The two-leg clock gate (consolidated)

**M47.W [DERIVED]:** CRT-4a  $\Leftrightarrow$  weight proportionality (Connes cocycle calculus). **M47.CI [DERIVED, shadow]:** the corner cocycle  $h^{\wedge}\{it\}$ , non-central. **M47.U (no-mixed-outcome) [DERIVED]:** tracial transport  $\Rightarrow$  parent  $\Pi_1$  + clock refuted; size-biased  $\Rightarrow$  parent  $R_\infty$  + clock survives — one binary, both branches pre-computed. **O2-a [DERIVED]:** the ZS-A24 mediator is microstate-democratic (uniform rate  $\Gamma_0\kappa^2$ ; doubly stochastic; stationary = trace) — the dissipative layer is structurally tracial and structurally unable to decide (O2-d). **O2-b [DERIVED, exact]:**  $\omega$  = the coincidence-conditioned two-leg law,  $\tau^2/\Sigma\tau^2$ , with  $\Sigma\tau^2 = 49/121$ . **O2-f (character weights) [DERIVED]:** with the sectors carrying the seam boost,  $\chi_d(\rho) = \sinh(d\rho)/\sinh \rho$ ; the fusion multiplicativity  $\chi_Y = \chi_Z \cdot \chi_X$  is exact; the two-leg law  $\rightarrow \omega = (4, 9, 36)/49$  at  $\rho \rightarrow 0$ , with L1 deviation 0.00919 at ZS-F30's bound  $|\rho_K| \leq \frac{1}{2}\ln(9/7)$ . **O2-g (parity selection) [DERIVED]:** the signed transport is  $\beta$ -odd ( $\beta_K\beta = g_K^{-1}$ , the definitional seam antisymmetry), the character weights are  $\beta$ -even — **gauge invariance selects the two-leg law**. **M47.T12 [DERIVED-CONDITIONAL]:**  $\beta$  = the  $2\pi$ -rotation spin- $\frac{1}{2}$  sign (ZS-A30 §8.2 verbatim; ZS-F23's subtitle), the deck involution in F30's half-angle coordinates ( $\varphi = 2\theta$ ); register action  $(-1)^{\wedge}\{2j\} = (-, +, -)$ , a fusion character, distinct from J (split (6,5)) and J\_Z (split (10,1)). **M47.HR [DERIVED, corpus-anchored]:** the X-leg is F30's spin-1  $\mathfrak{so}(3)$  axes; the Z-leg is F23's spin- $\frac{1}{2}$ . **M47.YM (mediation selection) [DERIVED-CONDITIONAL on HR + (MX)]:** the irreducible spin-5/2 Y-reading kills all three mediation vertices ( $\frac{1}{2} \otimes 1 = \frac{1}{2} \oplus 3/2$ ;  $\frac{1}{2} \otimes 5/2 = 2 \oplus 3$ ;  $1 \otimes 5/2 = 3/2 \oplus 5/2 \oplus 7/2$  — YL1–YL2), contradicting the founding axiom; the tensor reading  $Y \cong Z \otimes X$  keeps all alive (YL3), with the explicit  $6 \times 6$  lifts re-deriving the discriminant  $\Delta\chi = -0.392090$  at matrix level and the Casimir spectrum  $\{3/4, 15/4\}$  excluding  $35/4$  (YL4–YL5).

**The gate's verdict [DERIVED-CONDITIONAL on (MX) + (H1)]:** the transported corner weight is the size-biased  $\omega$  — exactly at  $\rho \rightarrow 0$ , within the computed sub-percent band at finite rapidity; the parent envelope on this branch is  $R_\infty$  and CRT-4a survives (M47.U). Handed to ZS-F37: the action-level test of (MX) and (H1), and the finite- $\rho$  consistency measurement (the predicted curve is  $\chi_2\chi_3$ , not  $\chi_6$ ).

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**§7. Theorem C [DERIVED-CONDITIONAL] —**  
nontriviality and  $\Omega$ -cyclicity of  $R = N' \cap M$  (case (A)  
automatic for second quantization); model identification  
 $R \cong A(0,1)$  with Hislop–Longo rates  $\pm 2\pi$ .

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## §8. NDC3: the charter delivered (consolidated)

**The instruments.** **M47.N1 [DERIVED-CONDITIONAL]:** the joint-a.c. spectral obstruction (necessity by counterexample). **M47.MP [PROVEN, group]:** the Möbius-closure no-go (the  $\mathrm{PSL}(2, \mathbb{R})$  centralizer of  $U(1)$  is the translation subgroup). **M47.GK/SIG:** the  $\mathrm{SL}(2, \mathbb{C})$  parabolic viability; the signature fork over the real forms of  $\mathfrak{so}(4, \mathbb{C})$ ; the condition-(2) obstruction on any single chiral line. **M47.CS/CS'/CS2**

**[DERIVED-CONDITIONAL on KH1–KH4]:** the conjugation-transfer theorem ( $\text{KH}\square(\tilde{f}) \Leftrightarrow \text{KH}\square(f)$  — the anti-chiral standard pair costs nothing new); the same-space no-go; the tensor two-seam Borchers triple with  $\hat{S} = \text{flip}^\circ(C \otimes C)$  and condition (4) at the operator level.

**The geometry. M47.HOL-1 [DERIVED, exact]:**  $\text{AdS}_3$  is the  $\text{SL}(2, \mathbb{R})$  group manifold (the det quadric; Killing signature  $(+, +, -)$ ; the left–right isometry action — HL1–HL4). **M47.HOL [DERIVED-CONDITIONAL on KH + (H-2D); Rehren IMPORTED-PROVEN]:** the two-seam net is a  $2+1$ -dimensional  $\text{AdS}_3$  wedge net.

**The operator. M47.D3-L [OBSTRUCTION, exact]:** the Lorentzian Dirac symbol dies on the light cone — no elliptic Weyl law on the Lorentzian group manifold. **M47.CD [DERIVED; Hardy–Ramanujan imported]:** the boundary spectra are a.c. (modular) or Cardy-growing (conformal,  $\log N \sim \pi\sqrt{(2\lambda/3)}$ ; exact partitions to  $p(200) = 3\,972\,999\,029\,388$ ) — **no boundary Weyl-3 operator exists; (H-cd) is a necessity by exclusion. M47.D3 [DERIVED-CONDITIONAL on KH1–KH4 + (H-2D) + (H-cd); spectrum IMPORTED-PROVEN]:** on the compact dual  $\text{SU}(2) \cong S^3 = \text{the unit quaternions}$  ( $\det q = a^2 + b^2 + c^2 + d^2$  — the Euclidean mirror of the  $\text{AdS}_3$  quadric), the Dirac operator has spectrum  $\pm(n + 3/2)$ , **multiplicity  $(n+1)(n+2)$** , with the exact counting  $\Sigma m(m+1) = K(K+1)(K+2)/3$  and

$$N(\lambda) = (2/3)\lambda^3(1 + O(1/\lambda)), \quad \lambda_{-n} \sim (3n/2)^{1/3}$$

— **the metric-dimension-3 Weyl growth of ZS-A17 Theorem F** (machine slope 2.9921). **The charter is delivered.** The ZS-A17 Spin–Metric NO-GO stands untouched (the geometry arrives through the group structure and imported theorems); the  $\text{AdS/dS}$  cosmological identification is NON-CLAIM; the Lorentzian bulk spectral theory remains with the registered toolset.

## §9. The Szegő gate: secured on KH1–KH4 alone — completed in v2.0

### §9.1 The gate (consolidated)

**M47.SZ [DERIVED reformulation]:** ZS-M46's KH4 and the  $(\star\star) c = 0$  corner are both "no singular inner factor" statements in the Beurling–Lax lattice of the seam Hardy space, whose prototype singular inner function is the seam translation  $e^{\{iaz\}}$  itself; both analytic cores are guarded by **(SZ):**  $\int \log w \, d\mu > -\infty$ .

### §9.2 The evaluation (consolidated from v1.8–v1.9)

**SZ-pos [PROVEN]:**  $f' = (i\pi/2)f$  never vanishes  $\Rightarrow$  the Koenigs derivative  $\chi' = \prod_{j \geq 1} f^{(j)}(z)/z^*$  (the exact product formula, ST-1) is zero-free on the basin — the interior-degeneracy mode cannot occur. **The measure [exact]:** the strip harmonic density is  $(1/(2\pi)) \operatorname{sech} t$  with first moment  $4 \cdot \text{Catalan}$ . **The tail rate (M47.ST) [DERIVED, exact]:** the exact ratio identity  $\chi'(z_{-k+1}) = (z_{-k}/z^*)\chi'(z_{-k})$  plus the period-3 Julia anchor ( $\text{GM} = 0.9015249$ ;  $f$ -multiplier 2.8398) give the **exact** backward-end rate

$a = \ln( GM|cycle| / |z^*| ) = 0.46274971$  per step (branch-uniform minimum),

whence  $\int \log w \, d\mu \geq -a \cdot (4 \cdot \text{Catalan}) / (2\pi) = -0.2698 > -\infty$ .

### §9.3 Theorem M47.GEO: (SZ-geo) discharged — NEW in v2.0

**Theorem M47.GEO [DERIVED].** *Any KH suspension — any map  $s$  from the basin to the strip that intertwines the seam dynamics with a nonzero real modular translation,  $s(f(z)) = s(z) + c_0$  — sends forward orbits to one strip end and backward orbits to the other:*

$\text{Re } s(f^n z) = \text{Re } s(z) + n \cdot c_0 \rightarrow \pm\infty$  (GEO1).

*The strip-end/branch-end identification is therefore not an additional hypothesis but the **defining property of the construction**: a map that fails it is not a suspension. The intertwining itself is verified mechanically at the Abel level: with the Koenigs value computed independently at each of thirty orbit points and the argument branch-unwrapped, the Abel increment is*

$u(f z) - u(z) = 1$  to  $3.3 \times 10^{-15}$  (GEO2),

*and the branch period is the exact locked-data constant  $\tau = 2\pi i / \log \lambda^* = 2.7743 - 0.1410i$  (GEO3).*

**Consequence.** The Szegő gate — and with it KH4's analytic core and the (★★) corner's analytic core — now rests on **KH1–KH4 alone**: the same registry every conditional statement of this paper already carries. The condition (SZ-geo) is retired (*gate F-M47.22 finalized*).

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## §10. The period-3 anchor: structure and the executed anti-numerology MC — NEW in v2.0

**The explicit cycle.** The principal backward attractor's three points have moduli (1.3413, 1.6574, 0.3296); the orbit is a genuine period-3 orbit of the entire map —  $f^3(z_1) = z_1$  to fifty digits (CY1) — with the chain-rule multiplier identity  $|(f^3)'| = (\pi/2)^3 \cdot \prod |z_i| = 2.8398$  exact (CY2).

**The anti-numerology MC [EXECUTED; protocol Step 5].** (i) *The scan*: against the twenty-six locked/derived corpus constants, the nearest neighbors of the two new numbers are  $a = 0.46275 \rightarrow \ln(\pi/2)$  at relative distance 2.4% and  $GM = 0.90152 \rightarrow |\lambda^*|$  at 1.1% — **no match at the corpus exactness bar (relative  $10^{-3}$ )**: the non-identifications of v1.9 are correct (CY3). (ii) *The Monte Carlo*: the probability that a uniform random value lands as close to *some* listed constant as ours did is  $p = 0.352 \gg 5\%$  (100 000 draws; CY4) — the observed proximities are statistically unremarkable, and per the protocol the correct action is **withholding all identifications**, which this paper does. (iii) *Family stability*: under  $\pm 2\%$  and  $\pm 5\%$  perturbations of the exponent coefficient, the backward period **remains 3** (CY5) — the anchor's period is a structurally stable feature of this dynamical family, **not** a fingerprint of the locked value  $\pi/2$ ; the period-3/dim-X coincidence-candidate therefore stays firewalled with quantitative justification



(F-M47.10/20). Status: the two firewalled observations of v1.9 are confirmed as non-identifications by executed MC — the strongest available anti-numerology outcome for a NON-claim.

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## §11. Zero-parameter and cross-version audit (protocol 3.1–3.3)

**3.1 Zero fitted parameters [PASS].** Inputs: the ZS-M1 germ ( $f = e^{\{i\pi z/2\}}$ ,  $z^*$ ,  $\lambda^*$ ); LOCKED  $A = 35/437$ ,  $Q = 11$ ,  $d = (2, 3, 6)$ ,  $\kappa^2$ ;  $2\pi$ ; ZS-F30's rapidity bound. Every other number is an exact evaluation (the sech measure, 4·Catalan, the cycle data, the CG tables, the Dirac spectrum, the partition values) — audited per-check across all ten suites (C2). The executed MC (CY4) closes the anti-numerology requirement for this paper's new constants.

**3.2 Cross-version dependency trace [PASS].** The  $z^*$  digits match ZS-M1/F31 App. D at every use (C3); upstream statuses are never reversed (C4); the own-paper retraction ( $v1.1 \rightarrow v1.2$ ) is documented (C5); every inter-version transition  $v1.2 \rightarrow \dots \rightarrow v2.0$  carries a non-reversal guard (C6–C13), with the sole superseded items being this paper's own strategies and conditions, each retired by theorem, not silently. ZS-M46's KH registry is consumed unchanged and is *strengthened* (KH1 machine-supported; KH4's analytic core secured); ZS-F30/F31/F23/A30/A17/A23/A24's objects are consumed verbatim.

**3.3 Observational non-collision [PASS].** No dimensionful prediction is made; Planck 2018  $\Lambda$ CDM values, Standard-Model couplings, and the frozen  $w = -1$  branch are untouched; the AdS<sub>3</sub> bulk carries the explicit NON-CLAIM against cosmological identification (F-M47.18); the Buckingham- $\pi$  no-go on absolute scales is inviolate (F-M47.7).

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## §12. Theorem M47, the condition trajectory, and the handover — the consolidation (v2.0)

### §12.1 Theorem M47 (Program Theorem 2 of 3)

#### [DERIVED-CONDITIONAL]

Under ZS-M46's KH1–KH4, the Z-Spin seam inclusion carries: (I) the canonical endomorphism  $\Gamma = U(2)$  with the T6 tunnel [DERIVED]; (II) faithful genuineness — no faithful normal conditional expectation (model-fixed), with T4-★ reduced to the seven-fenced (★★) whose analytic core is Szegő-secured on the same KH registry [DERIVED-CONDITIONAL]; (III) the two-leg clock: the size-biased corner weight  $\omega = (4, 9, 36)/49$  and the  $R_\infty$  parent branch, with every representation input corpus-anchored [DERIVED-CONDITIONAL on (MX) + (H1)]; (IV) the charter object: a Dirac operator with the exact metric-dimension-3 Weyl law on the unit-quaternion compact dual of the seam's own Möbius group, the boundary alternative being Cardy-closed [DERIVED-CONDITIONAL on (H-2D) + (H-cd)]. **The**

modular-depth line of ZS-A31 §6.3 is delivered at the conditional level, on a registry of eight named conditions of which four are ZS-M46's own.

## §12.2 The condition trajectory (CN1)

| Item                  | v1.x status       | v2.0 status                                    | Mechanism                                                     |
|-----------------------|-------------------|------------------------------------------------|---------------------------------------------------------------|
| M47.P / CRT-1'        | DERIVED (v1.1)    | <b>RETRACTED</b> →<br>Realization Problem      | external review + PE1                                         |
| (H-β)                 | hypothesis (v1.5) | <b>discharged</b>                              | deck derivation (v1.6) + A30 §8.2<br>textual anchor (v1.7)    |
| (H-rep)               | hypothesis (v1.4) | <b>discharged</b>                              | HR corpus anchoring (v1.8) + YM<br>mediation selection (v1.9) |
| (SZ-tail)             | condition (v1.8)  | <b>discharged</b>                              | the exact tail-rate theorem ST (v1.9)                         |
| (SZ-geo)              | condition (v1.9)  | <b>discharged</b>                              | the intertwining argument GEO (v2.0)                          |
| <b>Final registry</b> | —                 | <b>KH1–KH4; (H-2D);<br/>(H-cd); (MX); (H1)</b> | + open math: (★★)-localization,<br>(★_w), P2, (TT)            |

## §12.3 The handover (→ ZS-F37 / ZS-A32)

| Item                     | Recipient | Deliverable handed over                                                                             | Test                                                            |
|--------------------------|-----------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| (MX) action-level        | ZS-F37    | the CG-vertex form of the founding<br>axiom                                                         | derive the mediation vertex<br>from the F37 action;<br>F-M47.27 |
| (H1) + C_int             | ZS-F37    | the corner-compatibility and<br>iteration/modular-step matching<br>( $-\mu/\omega_Z = \theta/v_Z$ ) | the F31 App. D ratio                                            |
| finite-ρ consistency     | ZS-F37    | the predicted Y-character curve $\chi_2\chi_3$<br>(not $\chi_6$ ); $\Delta\chi = -0.392090$         | F-M47.24 (consistency<br>form)                                  |
| t* = Q depth<br>selector | ZS-F37    | the canonical tunnel (T4-E') with<br>no rung selected                                               | NON-CLAIM here                                                  |
| (H-2D) designation       | ZS-A32    | the conjugate seam ( $\lambda^*$ , equal<br>modular speed) + the tensor<br>Borchers triple          | KH-transfer theorem                                             |

| Item                               | Recipient                  | Deliverable handed over                                                    | Test             |
|------------------------------------|----------------------------|----------------------------------------------------------------------------|------------------|
| P2 envelope;<br>(★_w); (TT)        | ZS-A32                     | the three candidate constructions<br>E1/E2/E3; the weight-level<br>program | §5; §3           |
| Lorentzian bulk<br>spectral theory | ZS-A32                     | the registered vdD–P–R toolset; the<br>AdS <sub>3</sub> wedge net          | R15/R29-adjacent |
| (★★) localization<br>residual      | open<br>(M47-intern<br>al) | the seven-constraint fence + the<br>Szegő-secured core                     | F-M47.9          |

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## §13. Falsification gates (protocol 3.5; three tiers, consolidated)

**Tier 1 — mathematical / immediate rejection.** **F-M47.1:** a faithful normal state with  $\sigma^\wedge\varphi$ -invariant  $N$  refutes (★\_s). **F-M47.9:** a (★★)-intertwiner must defeat all seven constraints — maximally informative. **F-M47.2:** (UV)/KH2 failure voids T4-C. **F-M47.12/13:** exotic-envelope / abstract-(TT) failures. **F-M47.21/26:** the  $S^3$  spectrum-counting identity; the 3-cycle data and the ratio identity — immediate, exact recomputations.

**Tier 2 — structural / conditional.** **F-M47.4/5:** provenance (PASS); case (A). **F-M47.6 (final):** the clock gate rests on (MX) + (H1) + the sub-percent band; an F37 refutation of (MX) reopens the Y-leg. **F-M47.11:** route exhaustion (both NDC3 directions now closed: bulk delivered, boundary Cardy-obstructed). **F-M47.14/15/19:** character-law mismatch; the conjugate-seam KH transfer; the  $\beta$  textual anchor. **F-M47.22 (finalized):** the Szegő gate's exposure is now KH1–KH4 themselves — a KH failure voids the gate together with the entire program registry, as it should. **F-M47.25/27:** the Cardy scope; the (MX) axiom's explicit load-bearing role.

**Tier 3 — scope / numerology.** **F-M47.7:** Buckingham- $\pi$  (PASS). **F-M47.8:** two-tier separation — 148 exact/numerical vs 13 guards + 17 registrations (PASS). **F-M47.10/20 (finalized):** the fusion signature  $6 = 2 \cdot 3$ ; the three "2"s (conditional on the anchored spin- $\frac{1}{2}$  structure); the quaternion/ $4\pi$  resonance; the period-3/dim-X candidate — all registered, all firewalled, **all now backed by the executed MC ( $p = 0.352$ ): no identification is made, quantitatively justified** (PASS). **F-M47.18:** AdS/dS and Euclidean/Lorentzian companions (NON-CLAIM; PASS).

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## §14. Conclusion

Program theorem 2 of 3 is delivered. The seam inclusion's canonical endomorphism is its own doubled step; its genuineness is faithful in the precise sense that no normal expectation can average the seam's translation away; its clock weighs the register by the gauge-invariant two-leg law, landing on the size-biased  $\omega$  that makes the parent the injective  $\text{III}_1$  factor; and the geometry it generates — one Möbius group per seam, doubled by the conjugate seam, holographically a 2+1 AdS<sub>3</sub> net — carries, on its unit-quaternion compact face, a Dirac operator whose counting function is exactly cubic. The conditional registry ends at eight named conditions, four of them inherited unchanged from ZS-M46, plus four open mathematical items each carrying its decision procedure — and the paper's ten-version trajectory, with its one retraction, four discharged hypothesis chains, and two obstruction-forced placements, is retained in full as the honest record of how a conditional program should shorten its own assumptions.

The through-line, stated finally: **the Z-sector supplies the stage; the Z-Spin supplies every action on it** — the step  $U(1)$ , the sector  $U(2)$ , the transport  $g_K$ , the parity  $\beta$  (the spin- $\frac{1}{2}$  sign), the exchange  $\hat{S}$  — **the group of those actions is the stage's third dimension, its compact face carries the charter's operator, and the seam's own critical-point-free dynamics guards the analytic gate at the rate set by its own Julia anchor.**  $(A, Q, \dim Z) = (35/437, 11, 2)$  LOCKED.

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## §15. Claim ledger (v2.0, consolidated)

| #                                                                   | Claim                                                             | Status                                  |
|---------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------|
| T4-A / T4-B / T4-C                                                  | Takesaki reduction; Haagerup implication; product-state mechanism | PROVEN (C given (UV))                   |
| T4-D; (★ <sub>s</sub> ) model-fixed                                 | model closure, faithful normal CEs                                | DERIVED-CONDITIONAL                     |
| T4-E / T4-E'                                                        | $\Gamma = U(2)$ ; tunnel = T6 chain                               | DERIVED / DER.-COND.                    |
| T4-★ $\Leftrightarrow$ (★★); the seven-constraint fence (O1-a...j') | reduction + eleven rungs                                          | PROVEN rungs; closure OPEN              |
| M47.P / CRT-1' (v1.1)                                               | —                                                                 | <b>RETRACTED</b>                        |
| P1 / P2 / P3; CRT-1                                                 | realization problem                                               | DERIVED / OPEN / DER.-COND.; OPEN       |
| M47.U; O2-a/b/d/f/g; T12; HR; YM                                    | the two-leg clock chain                                           | DERIVED (scoped); YM DER.-COND. on (MX) |

| #                                                    | Claim                                                                                                                                               | Status                                                                            |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| clock verdict                                        | $\omega = (4, 9, 36)/49$ ; parent $R_\infty$ ;<br>CRT-4a survives                                                                                   | <b>DERIVED-CONDITIONAL</b><br>on (MX)+(H1)                                        |
| MP; GK; SIG; CS/CS'/CS2;<br>HOL-1/HOL                | the two-seam geometry; $\text{AdS}_3 = \text{SL}(2, \mathbb{R})$                                                                                    | PROVEN(group)/DERIVED/D<br>ER.-COND.                                              |
| D3-L; CD                                             | Lorentzian + boundary<br>obstructions                                                                                                               | OBSTRUCTION / DERIVED                                                             |
| <b>M47.D3 — the charter</b>                          | <b>compact-dual Weyl-3 Dirac:</b><br>$\pm(n+3/2), (n+1)(n+2), N \sim$<br>$(2/3)\lambda^3$                                                           | <b>DERIVED-CONDITIONAL</b><br>(KH + (H-2D) + (H-cd))                              |
| SZ; SZ-pos; ST-1/ST-2/ST;<br><b>GEO</b>              | the Szegő gate, secured                                                                                                                             | PROVEN / DERIVED (exact);<br><b>gate on KH1–KH4 alone</b><br>(NEW)                |
| <b>period-3 anchor + MC</b>                          | <b>cycle (1.3413, 1.6574, 0.3296);</b><br><b><math>f^3 = \text{id}</math>; <math>p = 0.352</math>; period stable</b><br><b><math>\pm 5\%</math></b> | <b>DERIVED (exact) +</b><br><b>MC-executed</b><br><b>NON-identification (NEW)</b> |
| <b>Theorem M47</b>                                   | <b>the delivery theorem (I)–(IV)</b>                                                                                                                | <b>DERIVED-CONDITIONAL</b><br>(NEW consolidation)                                 |
| $t^* = \mathbf{Q}$ ; absolute scale; $\text{AdS}/dS$ | —                                                                                                                                                   | NON-CLAIM                                                                         |

## §16. Verification (protocol two-tier separation)

Eleven companion scripts; the suites do **not** machine-certify the imported theorems (R1–R17).

**EXACT / NUMERICAL — 148/148 PASS.** *v1.0 (22): K1–K4, AW1–7, MB1–4, BM1–2, TK1–3, FG1, SP1.* *v1.1 (14): GE1–3, CII–6, MII–4, LW1.* *v1.2 (10): PE1–3, OIa-1/2, PF1/OB1, ORD1–2, NIC.* *v1.3 (12): OG1–3, OAI–3, OB2–3, MP1–4.* *v1.4 (17): OLI, OE1, OG4–5, CHI–7, SG1–4, CJI–2.* *v1.5 (13): CSI–5, BG1–4, OHI–4.* *v1.6 (12): HLI–4, HBI–4, OJI–4.* *v1.7 (13): D31–35, T21–24, SZ1–4.* *v1.8 (15): SII–6, HRI–5, CDI–4.* *v1.9 (11): ST1–6, YLI–5.* *All re-run.* **v2.0 (9): GEO1** — the intertwining sends orbits to the strip ends; **GEO2** — the Abel increment = 1 with independently computed  $\chi$  per point, branch-unwrapped, to  $3.3 \times 10^{-15}$ ; **GEO3** —  $\tau = 2\pi i / \log \lambda^* = 2.7743 - 0.1410i$  exact; **CY1** —  $f^3(z_i) = z_i$  (50-digit; cycle moduli 1.3413, 1.6574, 0.3296); **CY2** — the multiplier identity 2.8398; **CY3** — the nearest-constant scan (no match at relative  $10^{-3}$ ); **CY4** — the anti-numerology MC,  $p = 0.352$  (100k draws); **CY5** — period-3 family stability ( $\pm 2\%$ ,  $\pm 5\%$ ); **CN1** — the final registry bookkeeping (8 conditions / 4 open / 1 retraction / 4 discharged).

**CONSISTENCY — 13/13 executed guards.** C1 chronology DAG; C2 zero-parameter firewall; C3 cross-version digit match; C4 upstream non-reversal; C5 own-retraction documentation; C6–C12 inter-version non-reversal (v1.2  $\rightarrow$  v1.9); **C13 (NEW)** v1.9  $\rightarrow$  v2.0: zero reversals; three completions ((SZ-geo) discharged; the consolidation; the MC-executed non-identifications).

**REGISTERED — 17 imported-theorem conditions (R1–R17):** Takesaki; Haagerup; Connes III-classification; Connes injectivity; Haagerup III<sub>1</sub>-uniqueness; Araki–Woods (factor hypothesis explicit); Wiesbrock/Araki–Zsidó; Borchers; Doplicher–Longo; Buchholz; D'Antoni–Longo; GLW/Lechner–Longo/BGL; Rehren holography; Hitchin/Bär (the  $S^3$  Dirac spectrum); Beurling–Lax–Szegő–Privalov (+ the Lorentzian-triple toolset noted, not used); Hardy–Ramanujan; Milnor/Bergweiler (linearization surjectivity, Julia repellers). **External-review note:** the suite is executed and reported by the authors; independent re-execution is invited (scripts attached).

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## §17. Acknowledgements & Code Availability

The eleven scripts reproduce all 148 + 13 checks, including the executed anti-numerology Monte Carlo. This work used AI tools (Anthropic Claude) for corpus and external-literature search, cross-paper integration, symbolic/numerical verification, and drafting, under Kenny Kang's editorial direction. The v1.2 correction cycle incorporated an external technical review whose identification of the M47.P error is gratefully acknowledged. The author assumes full responsibility for all content.

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## §18. Appendix A — Deep-exploration (if-tree) record, v2.0

**Step 0 (long list, 6).** (1) (SZ-geo) via explicit reconstruction of the full KH4 suspension map (the conformal identification u-plane  $\rightarrow$  strip, including the branch-period geometry); (2) (SZ-geo) via the intertwining property alone (the definitional route); (3) the consolidation as a new stand-alone summary paper (ZS-M47S); (4) the consolidation as the v2.0 final version of this paper (in-place); (5) the period-3 structure via exact algebraic relations among the cycle points and  $z^*$ ; (6) the period-3 structure via the anti-numerology MC + family-stability protocol.

**Dropped:** (1) — dominated by (2): the full conformal reconstruction requires fixing data (the  $C_{\text{int}}$  matching ratio) that the ends-question does not need; the intertwining route decides the question with what every suspension must have, and (1)'s residue (the exact aspect-ratio matching) is precisely the  $C_{\text{int}}$  item already in the F37 handover — no duplication. (3) — rejected: a separate summary paper would fork the version history and violate the corpus's one-paper-one-ledger discipline; the consolidation belongs in v2.0's §12. (5) — attempted and terminated honestly: no exact algebraic relation among the three cycle points and  $z^*$  was found beyond the defining equations ( $f^3 = \text{id}$ ; the multiplier identity), and the MC (6)

then *quantified* that the numerical proximities are unremarkable — (5)'s null outcome is (6)'s input, not a silent drop.

**Step 1 (MECE issue list, 3, by impact).** **I1** = (SZ-geo) by intertwining {2, with 1's residue routed to the handover}; **I2** = the consolidation {4, with 3 rejected}; **I3** = the cycle structure + MC {6, with 5's null as input}.

**Step 2 (if-trees).**

**I1:** IF the ends-identification needs the full conformal data — **no**: any suspension intertwines,  $s(fz) = s(z) + c_0$  with  $c_0$  real  $\neq 0$ , so  $\text{Re } s(f^n z) = \text{Re } s(z) + nc_0 \rightarrow \pm\infty$  — forward orbits to one end, backward to the other, **by the construction's definition** (GEO1)  $\Rightarrow$  **M47.GEO [DERIVED]**; mechanical support: the Abel increment = 1 at  $3.3 \times 10^{-15}$  with independent per-point  $\chi$  (GEO2); the branch period exact (GEO3)  $\Rightarrow$  **(SZ-geo) discharged; the Szegő gate rests on KH1–KH4 alone.**

**I2:** IF the delivery is scattered across ten versions — consolidate: **Theorem M47 (I)–(IV)** stated once (§12.1); the condition trajectory tabulated (4 discharged chains, 1 retraction, final registry 8 + 4; §12.2, CN1); the F37/A32 handover fixed (8 rows; §12.3)  $\Rightarrow$  **the program theorem is a single citable object.**

**I3:** IF the cycle relates exactly to corpus data — **no exact relation found** (the null of route 5). THEN quantify: nearest constants at 2.4% and 1.1% — above the  $10^{-3}$  bar (CY3); MC  $p = 0.352 \gg 5\%$  (CY4)  $\Rightarrow$  per protocol Step 5, **identifications withheld — and the withholding is now MC-certified**; family stability: period 3 under  $\pm 5\%$  (CY5)  $\Rightarrow$  the "3" is generic in the family, not tuned to  $\pi/2 \Rightarrow$  the dim-X candidate stays a firewalled coincidence with quantitative justification. Positive content retained: the explicit cycle (CY1) and the multiplier identity (CY2) are exact structure for any successor.

**Step 3 (statuses).** I1: GEO DERIVED; (SZ-geo) discharged; the C\_int residue handed over. I2: Theorem M47 DERIVED-CONDITIONAL (consolidation); registries CN1-audited. I3: cycle structure DERIVED (exact); all identifications NON-CLAIM with executed MC ( $p = 0.352$ ); family stability CONSISTENCY.

**Step 4 (convergence).** Cycle 1 changed three node statuses relative to the v1.9 registry ((SZ-geo) discharged; the consolidation object created; the MC executed converting two pending firewalls into certified non-identifications). Cycle 2 changed **0**.  $0 < 3 \Rightarrow$  **converged**; MECE retained.

**Step 5 (scoring).** Against v1.9: the last hypothesis-flavored condition of the analytic gate is discharged by the *cheapest possible argument* (the construction's own definition); the ten-version ledger becomes one citable theorem with a condition-trajectory audit trail — the registry ends **shorter than the program began** (four of eight conditions are M46's own); and the anti-numerology protocol is executed rather than merely pre-registered, returning  $p = 0.352$  — the protocol's own criterion ( $p > 5\% \rightarrow$  **withhold**) is applied *against* the temptation to identify, the strongest possible demonstration of the discipline. Per the Step 5 rubric: **converged + corpus-non-colliding + anti-numerology MC executed with the withholding outcome** — the paper's registered non-claims are certified, and its conditional claims stand

on the shortest registry of the series. Verdict: **the program theorem is delivered; the version line closes at v2.0.**

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## §19. References

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**Internal (Z-Spin corpus):** ZS-M46 v1.5; ZS-M1; ZS-M2; ZS-A17 v1.5; ZS-A23; ZS-A24 v2.1; ZS-A30 v1.6; ZS-F23 v1.3; ZS-F30 v1.4; ZS-F31 v1.4; ZS-F32 v1.5; ZS-F33; ZS-A31 v1.5; ZS-A27 v2.3; ZS-A28 v2.0; ZS-F36 v2.1; ZS-F1; ZS-M6.

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## §20. Version History

**v2.0 (July 2026): The delivery release (Final).** Consolidates v1.0–v1.9 into Theorem M47 (I)–(IV) with the condition-trajectory table and the ZS-F37/ZS-A32 handover. New content: Theorem M47.GEO ((SZ-geo) discharged by the intertwining property; the Abel increment verified at  $3.3 \times 10^{-15}$  with independent per-point Koenigs values; GEO1–GEO3) — **the Szegő gate rests on KH1–KH4 alone**; the explicit period-3 structure (cycle moduli 1.3413/1.6574/0.3296;  $f^3 = \text{id}$  at 50 digits; the multiplier identity 2.8398; CY1–CY2) with the **executed anti-numerology MC** (no constant match at  $10^{-3}$ ;  $p = 0.352 \gg 5\%$ ; period stable under  $\pm 5\%$  family perturbation; CY3–CY5) certifying the non-identifications; the final registry bookkeeping (CN1). Verification  $22+14+10+12+17+13+12+13+15+11+9 = \mathbf{148/148}$ , guards 13/13, registrations 17. Zero fitted parameters;  $(\mathbf{A}, \mathbf{Q}, \dim \mathbf{Z}) = (35/437, 11, 2)$  LOCKED.

**v1.9 (July 2026): The decision release.** The tail-rate theorem (period-3 anchor;  $a = 0.46274971$ ); the mediation selection theorem ( $Y \cong Z \otimes X$ ). 139/139.

**v1.8 (July 2026): The evaluation release.** SZ-pos PROVEN; the sech/Catalan measure; HR anchoring; the Cardy closure. 128/128.

**v1.7 (July 2026): The charter release.** The compact-dual Weyl-3 Dirac operator; the  $\beta$ -identification; the Szegő gate. 113/113.

**v1.6 (July 2026): The dissolution release.**  $\text{AdS}_3 = \text{SL}(2, \mathbb{R})$ ; the deck derivation; the  $c = 0$  witness. 100/100.

**v1.5 (July 2026): The transfer-and-parity release.** The conjugation-transfer theorem; the tensor two-seam net; the parity selection. 88/88.

**v1.4 (July 2026): The character-decision release.** The character weights; the signature fork; the conjugate-seam candidate. 75/75.

**v1.3 (July 2026): The three-front narrowing release.** Dilation rigidity; the transport-route split; the Möbius no-go. 58/58.

**v1.2 (July 2026): The correction-and-unification release.** M47.P RETRACTED  $\rightarrow$  the Realization Problem; the no-mixed-outcome theorem. 46/46.

**v1.1 (July 2026): The if-tree execution release.**  $\Gamma = \text{U}(2)$ ; the tunnel; the corner dichotomy. Contained the retracted M47.P.

**v1.0 (July 2026): Initial public release.** The T4-full architecture; the V-C audit; the NDC3 frame. (Consolidated from internal Z-Spin Collaboration research notes up to v2.0.)